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A Short Introduction to Cytogenetic Studies in Mammals with Reference to the Present Volume

A. Graphodatsky[a] M.A. Ferguson-Smith[b] R. Stanyon[c]

[a]Institute of Molecular and Cellular Biology, Siberian Division of the Russian Academy of Sciences, Novosibirsk, Russia; [b]Department of Veterinary Medicine, University of Cambridge, Cambridge, UK; [c]Department of Evolutionary Biology, University of Florence, Florence, Italy

Key Words
Chromosome painting · Eutherian ancestral karyotype · Genome evolution · Phylogenomics

Abstract
Genome diversity has long been studied from the comparative cytogenetic perspective. Early workers documented differences between species in diploid chromosome number and fundamental number. Banding methods allowed more detailed descriptions of between-species rearrangements and classes of differentially staining chromosome material. The infusion of molecular methods into cytogenetics provided a third revolution, which is still not exhausted. Chromosome painting has provided a global view of the translocation history of mammalian genome evolution, well summarized in the contributions to this special volume. More recently, FISH of cloned DNA has provided details on defining breakpoint and intrachromosomal marker order, which have helped to document inversions and centromere repositioning. The most recent trend in comparative molecular cytogenetics is to integrate sequencing information in order to formulate and test reconstructions of ancestral genomes and phylogenomic hypotheses derived from comparative cytogenetics. The integration of comparative cytogenetics and sequencing promises to provide an understanding of what drives chromosome rearrangements and genome evolution in general. We believe that the contributions in this volume, in no small way, point the way to the next phase in cytogenetic studies.

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Cytogenetic analysis of mammalian karyotypes has gone through a number of phases strictly related to technical advances. Hsu, in his entertaining book entitled 'Human and Mammalian Cytogenetics: an Historical Perspective', referred to the earliest phases as the 'dark ages' and the scientists who pursued chromosome studies were quaintly deemed 'knights' [Hsu, 1979]. Then came the 'hypotonic miracle' of the 1950s, which culminated in deciphering the correct human diploid chromosome number. It is a curious fact that diploid numbers of various mammals, including relatives of man, such as the chimpanzees and macaques, were already correctly described by this time. Yet hypotonic treatment along with tissue culture and the discovery of mitogens led to an explosion in papers on the diploid number and chromosome morphology of literally hundreds of mammals. More than 400 of these karyotypes were published in 'An Atlas of Mammalian Chromosomes' edited by Hsu along with Benirschke as a multi-volume set [Hsu and Be-

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Roscoe Stanyon
Department of Evolutionary Biology
University of Florence
IT–50125 Florence (Italy)
roscoe.stanyon@unifi.it

nirschke, 1967–1991]. On the basis of classical cytogenetics it was easily shown that different species had different number and form of chromosomes, i.e. they had different karyotypes. Karyotypes could vary greatly even between related species, and hypotheses about ancestral genomes were framed in terms of the diploid number (2n) and the number of chromosome arms (fundamental number, FN). We now know that mammalian chromosome numbers range from 2n = 6 in the Indian muntjac to 2n = 102 in the viscacha rat.

Just when classical studies of chromosomes seemed somewhat in a stall, then of course came chromosome banding, beginning with Q- and going on in short order to G-, R- and C-banding and many others, fluorescent and non-fluorescent. The immediate utility of banding was matching homologous chromosomes of the same cell and describing in more detail translocations and inversions in human clinical cases. In no time at all these technical advances were applied to animal species and in particular the mammals. Again an explosion of reports was forthcoming. Literally, hundreds of species were and are still being described at this level. Workers began looking for between-species homologies. These were most obvious between closely related species such as humans and their primate relatives, the great apes and Old World monkeys, with some notable failures such as the lesser apes. Even though it was clear that homology was proposed essentially on the basis of morphological similarity, bolder individuals looked even further afield and matched human chromosome banding with that of other mammals such as cats [O'Brien and Nash, 1982] and rabbits [Dutrillaux et al., 1980]. From these banding era comparisons it was rightly concluded that large blocks of mammalian genomes were conserved over millions of years. A curious fact almost ignored was that, even though the conclusions were right, about 50% of the homologies based on banding proposed between human/cat or human/rabbit was shown to be wrong in the chromosome painting era. About the same percentages were found in early banding homologies between macaques and green monkeys [Stock and Hsu, 1973].

We must conclude that establishing chromosomal homology between distantly related species or species characterized by rapid chromosomal evolution remained speculative until the advent of molecular cytogenetics and especially chromosome painting in the late 1980s and early 1990s [Wienberg et al., 1990].

Of course, by that time in situ hybridization was nothing new. It had begun even before the advent of banding with the publication of Gall and Pardue in 1969 [Gall and Pardue, 1969]. These early techniques have much in common with those of today. The chromosomal DNA on the slide and the probe both needed to be denatured. However, most molecular cytogeneticists now have little appreciation of how time-consuming and difficult these techniques were. The probe had to be radioactively labeled and after hybridization the slides were covered with a photograph emulsion. After sometimes weeks of exposure, the photograph emulsion was developed. Except for the most highly repetitive parts of the genome, the autoradiographs were difficult to analyze. The radioactive in situ hybridization procedure made the normal staining, microphotography, film development, printing and manually cutting and assembling karyotypes seem like a breeze (all procedures notably shortened today with the advent of digital imaging). Modern molecular cytogenetics was really born only with the discovery of nonradioactive methods [Langer et al., 1981] and probes, which initially came out of the human genome project. These initial hybridization techniques relied on biotin labeling and detection with avidin coupled with a fluorescent reporter molecule (fluorescence in situ hybridization, FISH).

A History of Chromosome Painting

Early hybridization of whole chromosome-specific probes (whole chromosome paints, WCP) used plasmid libraries and nick translation. This was sometimes termed CISS (chromosomal in situ suppression) hybridization [Cremer et al., 1988], due to the use of Cot-1 DNA to block the repetitive fraction of the genome from participating in the in situ hybridization. Later the easier, more visual 'chromosome painting' [Pinkel et al., 1988] became the exclusively used term. Soon after, chromosome painting began to establish its utility for both medical analysis and comparative research. Wienberg et al. [1990] were the first to show that chromosome painting could be used to compare genomes. Studies soon followed which mapped the chromosomal homology of entire genomes [Jauch et al., 1992; Wienberg et al., 1992]. The chromosome painting procedure based on plasmid libraries was still expensive and time-consuming, and, more importantly, it was also limited by the amount of probe DNA available which was rapidly consumed by the nick translations.

Then Carter et al. [1992] and Telenius et al. [1992] introduced the use of a degenerate oligonucleotide-primed PCR (DOP-PCR) for amplifying chromosome sorts. The

application of DOP-PCR methods using a random or universal primer made chromosome paints from flow-sorted chromosomes available almost without limit. The 6MW primer reported in these papers became the standard primer for the PCR of chromosome sorts. However, alternative random primers are possible such as the FS primer for mouse chromosomes [Rabbitts et al., 1995] and many others. Note that these whole chromosome paints should not be referred to as libraries, a frequent error even today. Another advantage was that it soon became clear that the quality of these paints was much improved and had vast untapped potential to chart genomes even between distantly related species [Scherthan et al., 1994]. A number of modifications were necessary for long-range phylogenetic painting. Among these was an increase in the amount of probe used and an increase in hybridization times, from several days to a week. Pretreatment of the metaphase chromosomes was also important both to improve target DNA disposability and better banding.

As is well known, chromosome paints rapidly became indispensable for a wide range of clinical and comparative research [Ferguson-Smith, 1997]. Part of this popularity was due to the fact that not only the paints were better, but also were cheaper and became more widely available. Chromosome painting was rapidly shown to be superior to chromosome banding for uncovering chromosome rearrangements in clinical cytogenetics and in reconstructing ancestral associations in comparative research. The results were much more reliable because homology is established not on the basis of morphological similarities, but on the basis of DNA content.

The laboratory of Ferguson-Smith became the center of an important cytogenetic industry: the comparison and mapping of between-species chromosomal homology no longer at the morphological level, but at the DNA level. It is difficult to find a comparative cytogeneticist working today who in some way, directly or indirectly, is not connected to what took place in the Cambridge laboratory on Tennis Court Road.

Flow-sorting and PCR introduced other additional possibilities to comparative cytogenetics. Chromosome paints could be produced from virtually any species for which an adequately dividing cell culture could be established. The possibility to flow-sort and establish chromosome paint sets from many species made reciprocal painting [Wienberg and Stanyon, 1997] and even multidirectional painting possible. Such reciprocal and multidirectional chromosome painting allows a much more precise inference of breakpoints involved in chromosome rearrangements. Literally hundreds of species were flow-sorted at Cambridge, many of which were distributed to researchers over the world through the Cambridge Resource Centre for Comparative Genomics which was funded by the Wellcome Trust [Ferguson-Smith et al., 2005]. Later a flow-sorting lab was established by Wienberg and Stanyon at the National Cancer Institute, Frederick (USA), and a second, more limited source of chromosome paints then became available.

Spectral Karyotyping, M-FISH, Chromosome Bar Coding

Additional specialized applications based on chromosome sorting and painting were also developed. The most spectacular of these is the staining of each of the 24 human chromosomes in different colors at the same time using whole chromosome painting probes. One method is known as spectral karyotyping, SKY [Schrock et al., 1996], and another is M-FISH [Speicher et al., 1996]. Although the results are fairly comparable, the methods are diverse. They require different combinations of filters and imaging systems. Another method is chromosome bar coding, which 'bands' chromosomes [Müller and Wienberg, 2001]. All these techniques resort to pools of chromosome paints. Although the resulting images are striking, all have had little impact in comparative molecular cytogenetics. Apparently the rapidly decreasing hybridization efficiencies in the long-range phylogenetic FISH make these methods unproductive except in the most closely related species. Therefore, we will not dwell further on these methods.

Microdissection and FISH with Cloned DNA

Chromosome paints are the largest probes used in molecular cytogenetics. Chromosome paints are very good at documenting interchromosomal rearrangements (translocations), but rarely provide information on intrachromosomal rearrangements (e.g. inversions). These limitations could be overcome by physically isolating a slice of a chromosome by microdissection [Guan et al., 1992] or by cloning DNA in various vectors such as YACs and BACs [Boyle et al., 1992; Kirsch et al., 2000]. Microdissection probes can be pooled for multicolor banding hybridizations [Liehr et al., 2006], which have had some limited application in both primates and rodents. BAC-FISH has become somewhat more popular for establishing marker order along chromosomes and investigating intrachromosomal rearrangements. One surprising results was the discovery of evolutionary new centromeres [Rocchi et al., 2012] in a number of diverse mammalian

orders. The use of BAC-FISH to determine marker order and centromere repositioning is also reviewed in this volume in the article by Stanyon et al.

Search for Ancestral Genomes

The reconstruction of the history of life on our planet is one of the fundamental goals of evolutionary biology. Darwin first showed that evolution operated through natural selection and descent from common ancestors. Darwin's theory that all forms of life on this planet are connected by descent from common ancestors is now an undisputed fact. This fact means that all forms of life on our planet are related and all species have common ancestors.

Since the days of classical staining, cytogeneticists have been interested in establishing the content of ancestral karyotypes. In discussion of ancestral karyotypes in the classical staining era, cytogeneticists often fell into one camp or another. Many students of chromosomes thought that the ancestral mammalian karyotypes had a high diploid number and that evolution predominantly proceeded by chromosomal fusions. Others hypothesized that fissions were the main evolutionary mechanism and proposed low ancestral diploid numbers while others took the middle ground, suggesting that both mechanisms were important.

An intriguing summary of these classically stained chromosome data is found in the paper by Matthey [1972] just at the beginning of the next great phase of cytogenetics, chromosome banding. Matthey, who among other things invented the term 'fundamental number', held the view that chromosome number could go up or down and therefore proposed that the ancestral placental mammalian karyotype should be close to the median of the 1,063 mammalian species he took into consideration. He decided on 48 chromosomes, surprisingly in line with modern chromosome painting hypotheses. In the same paper he discussed the karyotypes of marsupials and presented 2 hypotheses: an ancestral diploid number of 14 or 22. In the end he settled on 22 as the most probable ancestral diploid number for marsupials. The fascinating story of how molecular cytogenetics has finally allowed us to know that the ancestral number is more likely 14 is told in the contribution of Deakin et al. in this volume (see their figure 2). These authors also discuss why the ancestral chromosome number of monotremes, 2n = 46, is much higher than the 2n = 14 ancestral karyotype of marsupials and why the low diploid number is specific for the marsupials, not for all non-placental mammals. They then go on to derive many of the karyotypes in these species.

Ancestral karyotypes were also proposed for various points on the mammalian tree on the basis of banded chromosomes. Foremost among these were those of Dutrillaux and coworkers who proposed ancestral karyotypes for primates [Dutrillaux, 1979, 1988; Dutrillaux and Couturier, 1981] and carnivores [Dutrillaux and Couturier, 1983]. For example, they suggested an ancestral diploid number of 2n = 54 for the New World monkeys, the same number as discussed by de Oliviera et al. in this issue, but with some differences in content. Ancestral karyotypes, based on painting data and occasionally BAC-FISH and supported by gene sequencing data, are presented in a number of the reviews in this special issue. Perelman et al. thoroughly discuss the various hypotheses for ancestral karyotypes in the carnivores and Rubes reviews the progress in determining the ancestral karyotype of the Cetartiodactyla.

Trifonov et al., also in this issue, discuss the utility of gene mapping data in reconstructing the ancestral karyotype of the Perissodactyla and higher-order branching in placental mammals. In their comparison of cytogenetic data for Sciuromorpha and Lagomorpha, Richards and Dutrillaux hypothesize that uncertainties in determining the content of ancestral genomes can inform us about the tempo and mode of speciation in these taxa. Their conclusions would be interesting to test in a broad array of mammals.

The review of Nie in this volume also discusses the content of ancestral genomes and the various landmark rearrangements identified as characteristic for each different lineage, that potentially unite certain lineages within Strepsirrhine primates, Dermoptera and Scandentia. Finally, a catarrhine ancestral karyotype is also presented by Stanyon et al., which includes the probable marker order and centromere positions.

Tenets of Ancestral Genome Reconstruction

Establishing the ancestral eutherian karyotype has been one of the aims of interspecies chromosome painting studies. As discussed above, molecular cytogenetics allows great confidence in mapping chromosome homology between species. We can determine how many chromosome rearrangements are needed to transform the genome of one species into another. It is another but related task fundamental to phylogenetic reconstruction, to determine what chromosomes were present in ancestral species. Cladistic analysis of evolutionary relationships

depends on the polarity of chromosome rearrangements. Two species can be very similar to each other only because they have both retained conserved, ancestral chromosomes. Other species which have very different karyotypes can at the same time be linked by common derived rearrangements. Hypotheses on the content of ancestral genomes aid in recognizing conserved, primitive characters versus derived rearrangements.

Some basic tenets of constructing ancestral genomes from chromosome painting data were summarized by Wienberg and Stanyon [1995, 1997, 1998]. A chromosomal synteny may often be disrupted independently by chromosomal rearrangements, but it is highly unlikely that the same syntenic group was brought together independently in different lineages, especially in the same order. Parsimony analysis is then commonly used to determine ancestral chromosomal syntenies (the largest linkage group known) and associations. Commonality, combined with the inclusion of appropriate 'outgroups', helps determine if a chromosome is syntenic and what associations were present in an ancestral genome. An outgroup is one or more species from the next phylogenetic branch out from the group under discussion. For instance, the best outgroup for great apes and humans would be the Old World monkeys.

When a chromosomal synteny is found intact between various species, this condition is likely to be ancestral (symplesiomorphic). For instance, a chromosomal synteny found in widely divergent mammalian orders would suggest that it was present in the ancestor of all mammals. In the early stages of reconstructing the ancestral genome of living placental mammals, it became clear that all chromosomes homologous to human chromosome 3 and human chromosome 21 were syntenic in the ancestor [Müller et al., 2000]. This syntenic chromosome association, written as 3/21, was later disrupted in the evolutionary line leading to humans. Today it is not found in any Old World monkeys, apes or humans, but it is found in New World monkeys, prosimians and is universal in almost all other mammals, including marsupials, and even in chickens. Therefore, the fission of 3/21 would qualify as a phylogenetic landmark linking Catarrhine primates. Most authors appeal to the concept of parsimony and consider that likely ancestral chromosomes are commonly present in species of a number of divergent eutherian orders [Svartman et al., 2004].

The Ancestral Karyotype of Placental Mammals

Over the last few years various hypotheses were advanced on the content of the ancestral karyotype of all living placental mammals. The diploid number in these hypotheses ranges from 44 to 50 chromosomes [Svartman et al., 2004]. Wienberg and Stanyon [1997] were the first to formulate hypotheses about the probable content of the mammalian ancestral karyotype based on chromosome painting data, but it was Chowdhary et al. [1998] who published one of the first complete reconstructions. They proposed a diploid number of 48, with chromosomes 4, 7, 8, 10, 16p present as single chromosomes, and that the association 12/22a with an additional chromosome 22b was also likely part of the eutherian ancestral karyotype. However, with the inclusion of improved chromosome painting data from further taxa, it was deemed likely that chromosomes 2pq, 2q, 3/21, 5, 6, 4/8p, 7p, 7q/16p, 8q, 9, 10p, 10q, 11, 12q/22a, 12p/22b, 13, 14/15, 16q/19q, 17, 18, 20, X and Y were syntenic in the ancestral eutherian karyotype [Murphy et al., 2001; Richard et al., 2003]. An unsettled question was the possibility that 10/12q/22a might be present as an ancestral placental mammal association and there were some doubts whether chromosome 1 might be a single syntenic unit in the ancestor. Then, Murphy et al. [2003] demonstrated through comparative gene mapping data and reciprocal chromosome painting with homologs to human (HSA) chromosome 1 in 7 species from 6 eutherian orders of mammals that the segments previously considered equivalents of HSA1p and 1q in these species were not identical. In the meantime, a single HSA1 homolog was also found in 3 highly divergent eutherian orders. These data strongly indicated that a single large chromosome, corresponding to HSA1, was present in the ancestral eutherian karyotype [Murphy et al., 2003]. The contribution of Svartman and Stanyon in this issue discusses the 2 hypotheses in regard to the association 1/19. Due to its presence in Afrotheria, the 1/19 association was considered a possible candidate for inclusion in the ancestor of all placental mammals [Yang et al., 2003]. The alternative hypothesis was that the 1/19 association was instead a derived trait linking Afrotheria [Frönicke et al., 2003; Svartman et al., 2004; Liu et al., 2011]. As discussed by Stanyon and Svartman in this issue, the 1/19 and the 5/21 associations are both derived in Afrotheria. Again in this volume, Svartman suggests that the xenarthran ancestral karyotype has conserved better and is closest to the ancestral karyotype of all placental mammals. It can be noted that the 2 supposed xenarthran synapomorphic associations HSA2/8 and 7/10 [Yang et al., 2006; Liu et al., 2011] were also detected in the marsupial and chicken genomes [Graphodatsky et al., 2011]. Therefore, Xenarthra may be a better candidate for, or at least represent more

closely, the basal placental mammal. Clearly, this possibility deserves further study, both with further chromosome painting and sequencing.

Questions also arose whether there were conflicting results from other levels of genome analysis [Froenicke et al., 2006]. In a paper in Science, Murphy et al. [2005], while confirming many of the conclusions from molecular cytogenetics about the boreoeutherian ancestral genome, proposed 5 additional syntenic associations that had apparently gone undetected by chromosome painting. However, later more refined methods and additional data [Ma et al., 2006] gave no support to these additional associations [Rocchi et al., 2006].

Chromosome Homology and Evolution in Mammals, Birds and Reptiles: Hypothesis on the Role of Non-Coding DNA

(Section contributed by M.A.F.-S.)

While animal karyotypes display a great diversity in number and morphology of chromosomes, their genomes are remarkably conserved. This conservation is resolved by chromosome painting into small numbers of large chromosome blocks. Rearrangement of these blocks into different combinations explains much of the diversity in karyotypes. Evolutionary relationships of representative species of all mammalian orders can be determined from studying the pattern of associations of these syntenic blocks [Ferguson-Smith and Trifonov, 2007]. For example, the association of blocks from HSA3p and 21 can be found in the karyotypes of all mammals except the apes and Old World monkeys, as are blocks from HSA14 and 15. Other associations are signatures that characterize 1 group of species, for example the HSA5/21 association in Afrotheria. The chromosome-specific DNA used to delineate these homologies is prepared from flow-sorted chromosomes, amplified and labeled by PCR.

Unfortunately, the success of chromosome painting in determining chromosome homologies does not extend to all vertebrates. It is unsuccessful *between* placental mammals and marsupials or monotremes, or birds, or reptiles, although it works well enough *within* each of these groups to postulate ancestral karyotypes for each.

Except for part of the X chromosome [Glas et al., 1999], homologies cannot be demonstrated by painting between human and marsupials, although cross-species painting is effective in determining chromosome homology between all extant marsupials [Rens et al., 1999, 2001, 2003]. The chromosomes of egg-laying monotremes, which are the most basal clade of mammals and have a unique multiple X and Y sex chromosome system [Rens et al., 2004, 2007], also cannot be painted by human paint probes, although painting between platypus and echidna has been very informative [Rens et al., 2007]. Where cross-species painting fails to show homologous regions in more distantly related taxa such as chicken and platypus, recourse has been made to the more time-consuming gene mapping comparisons [Rens et al., 2007].

Marsupials and monotremes diverged from the mammalian lineage much earlier than placental mammals, approximately 148 and 166 million years ago (Mya), respectively [Bininda-Emonds et al., 2007]. This suggests that successful painting might be limited to taxa arising from a common ancestor no earlier than about 100 Mya. Thus cross-species painting would not be expected to work between mammals and birds. Painting chicken chromosome 4q to HSA4 is the only apparent exception to this general rule [Chowdhary and Raudsepp, 2000], although gene mapping and sequencing confirms that there are blocks of conserved homology between their chromosomes.

The chicken karyotype has served as the model for comparative chromosome studies in birds. It has 9 pairs of easily recognized macrochromosomes, plus the Z and W sex chromosomes, and 29 pairs of microchromosomes, many of which sort together in the flow karyotype and are difficult to characterize. Some microchromosomes can be isolated from single chromosome sorts or by microdissection and others rely on BAC mapping for identification [Griffin et al., 1999; Masabanda et al., 2004]. Macrochromosome probes have been used almost exclusively in comparative painting studies in over 50 species from at least 10 different avian orders. The main findings are extensive chromosome conservation in birds with a comparatively small number of interchromosomal rearrangements in those species with diploid numbers from 78 to 92. Raptors tend to have smaller chromosome numbers, from 48 to 66, resulting from fusions of ancestral macro- and microchromosomes. The stone curlew (*Burhinus oedicnemus*, 2n = 42) has one of the smallest chromosome numbers with multiple fusions of chicken microchromosomes [Nie et al., 2009]. The chicken karyotype is thought to be close to the ancestral avian karyotype.

Chromosome paints have been made recently in the Cambridge laboratory from several reptiles, including lizards *(Scincus scincus, Hemidactylus turcicus, Hemidactylus platyurus, Anolis carolinensis* and *Gekko japonicus)*,

turtle *(Trachemys scripta elegans)* and crocodile *(Crocodylus niloticus)*. Cross-species painting between these species has been very informative, although the hybridization efficiency has been less than in birds or mammals: this is understandable in view of the longer divergence times. The results are consistent with greater genome conservation and even fewer evolutionary rearrangements than in birds [Giovannotti et al., 2009; Trifonov et al., 2011].

In view of the failure of chromosome painting to demonstrate homologies between taxa that have diverged more than 100 Mya, recent results on successful painting between birds and reptiles were unexpected. The chicken Z probe was found to hybridize without apparent rearrangement to an autosome in all the above reptilian species despite over 250 million years divergence [Pokorna et al., 2010]. This observation prompted experiments with chicken macrochromosome probes onto reptilian chromosomes. Extensive homology was discovered between chicken, turtle and crocodile chromosomes [Kasai et al., 2012b] and between chicken and squamate species [Pokorna et al., 2012]. Thus, painting between chicken and turtle showed that chicken macrochromosomes 1–3 and 11 were each represented by a single turtle chromosome, and the arms of chicken 4, 6 and 8 were homologous to turtle 5 and 7p, 7q, and 8q, respectively. However, painting between chicken and crocodile revealed fission of 2 ancestral avian chromosomes with rearrangement by centric fusion into 4 crocodile chromosomes, contributing to the reduced chromosome number characteristic of the order. High chromosome conservation is thus a feature of birds and reptiles, and the results already suggest that the presence of ancestral avian chromosomes extends into at least the early Jurassic era.

The ability of cross-species chromosome painting to demonstrate chromosome homology between some vertebrate groups and not others is of some interest. The usual explanation is that chromosome painting involves hybridization of protein-coding sequences, and presumably also of regulatory sequences, to their complementary sequences on the chromosome. Painting signals tend to be more extensive the closer the relationship between 2 species. Its success must depend on the level of conservation, and thus the length of time elapsed since the divergence of the 2 species from a common ancestor. Repetitive sequences can be excluded because they are usually not specific to one particular chromosome and are suppressed during the painting procedure. Although vertebrate protein-coding sequences tend to be highly conserved, many human cloned genes do not hybridize easily to their homologues by FISH mapping, even in closely related species. For success the orthologue of each gene must be found and used. Also, protein-coding sequences represent less than 2% of genomes and seem an unlikely explanation for the depth and extent of hybridization signals observed across homologous regions.

An alternative hypothesis is that painting does not depend on gene sequence but on an as yet undefined functional class of chromosome-specific non-coding DNA (CSNCD) in greater abundance than protein-coding sequences. CSNCD could have evolved to ensure chromosome synapsis in meiosis and to reduce illegitimate recombination between non-homologues. Different levels of conservation of CSNCD can account, for example, for the success of painting between chicken and reptiles and its lack of success between human and marsupials. In other words, CSNCD in marsupials is expected to have diverged substantially from humans, as cross-species painting is ineffective between them, whereas the chicken CSNCD is shared to a considerable extent with reptiles. New chromosomes resulting from rearrangement during evolution will still maintain CSNCD, albeit in a different combination.

A recent study on the proportion of the genome that is functional suggests that 6.5–10% of the human genome is under functional constraint, and similar proportions are found for rat and mouse genomes [Meader et al., 2010]. These authors demonstrate in many pairs of species that the amount of functional DNA shared between 2 species decreases as the divergence between them increases. So far, the analysis has not extended to chromosome-specific DNA where one predicts that the proportion under functional constraint could be much higher.

Other factors could contribute to the painting results between chicken and reptiles. Compared to humans, chicken, turtles and crocodiles have much smaller genome sizes (between 1.0 and 2.0 Gb) [Kasai et al., 2012a] and perhaps have not accumulated those DNAs that tend to diverge from the more conserved DNA classes in animals with larger genomes. Also, there is evidence that chicken and reptile genomes share a higher percentage of long conserved non-coding sequences with one another than with mammals [Janes et al., 2011]. Nonetheless, the CSNCD hypothesis remains the most plausible mechanism to explain the observations obtained by cross-species chromosome painting in vertebrates. The hypothesis may be testable by comparative sequencing of chromosome-specific DNA when suitable tools are developed to interrogate genome databases, so that shared CSNCD can be distinguished from diverged CSNCD.

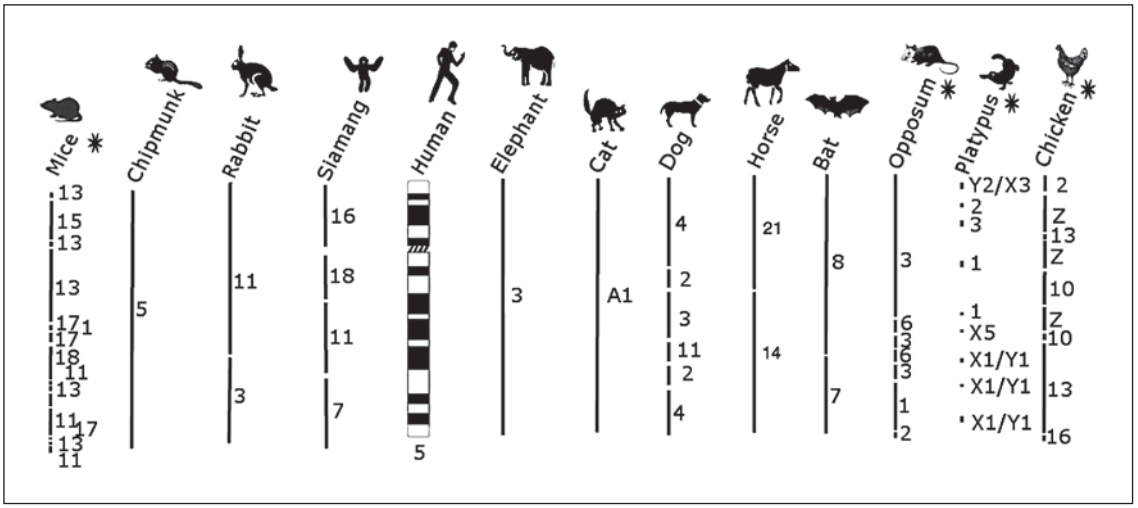

Fig. 1. Examples of homology between human chromosome 5, various mammalian species and the chicken. Placental mammals were compared on the basis of chromosome painting, while species marked by an asterisk were compared on the basis of sequence alignments (http://www.ensembl.org/) without Zoo-FISH data.

The Importance of Marsupials, Monotremes and other Vertebrates for Understanding the Evolution of the Placental Genome

In this issue, Redi and Capanna present a fascinating contribution on the relationships between genome size, taxonomic divisions and molecular mechanisms. These are very basic facts of genome structure and composition, which support the higher taxonomic divisions because they appear to vary among the superordinal clades. They also raise the good possibility, in line with the above speculations of M.A.F.-S., that new classes of DNA apparently raised the genome size of placentals, which inhibit good signals from placental paints on marsupials and monotremes.

As can well be appreciated from a causal glace at this volume, genome-wide comparative chromosome maps between humans and representative species of all extant eutherian orders have been established (fig. 1). However, the lack of good results using placental painting probes on marsupials and other vertebrates is not just an interesting theoretical question; it points out a critical fact. We do not have appropriate outgroups necessary to fulfill a fundamental requirement of cladistic analysis. As ably discussed above and in this volume by Deakin et al., there is outside of some hybridization data from the X chromosome a total lack of comparative chromosome painting data between eutherian and other mammals: monotremes and marsupials as well as other more distantly related vertebrates such as birds and reptiles. The only alternative at this moment is to compare the chromosome painting results to sequence assemblies (fig. 2).

Wang et al. [2011] and Deakin et al. (this volume) discuss how it is possible to make a virtual map of the chromosome homologies between marsupials and placentals. Other researchers have demonstrated how the genome sequence data can be used to electronically paint chromosomes. Initial results are quite informative [Kemkemer et al., 2006].

These comparisons show that at the time of the ancestral placental, mammalian associations (4/8, 10/12/22, 7/16, 16/19) are present in other vertebrates, lending good support to the hypothesis that they were also present in the ancestral placental genome. A 1/19 association is indeed also present in *Monodelphis domestica*, but this association is 1p36/19q13 while that in the Afrotheria is 1p/19p as discussed in the contribution by Svartman and Stanyon in this issue (see their figures 1 and 2).

We decided it might be worthwhile to compare the associations present in some non-placentals for which genome assembly data are available: the gray short-tailed opossum (*M. domestica*, MDO), the tammar wallaby (*Macropus eugenii*, MEU), the platypus (*Ornithorhynchus anatinus*, OAN), the chicken (*Gallus gallus*, GGA) and the green anole lizard (*Anolis carolinensis*, ACA). We are well aware that there are limitations based on various methods utilized to determine homology with human chromosomes and the diverse levels of coverage of the genome as-

Fig. 2. Genome-wide chromosomal correspondence among mammalian and avian species with the human ideogram as the reference. ECA = Horse *(Equus caballus)*, GGA = chicken *(Gallus gallus)*, HSA = human, MDO = opossum *(Monodelphis domestica)*, OAN = platypus *(Ornithorhynchus anatinus)*, and OCU = rabbit *(Oryctolagus cuniculus)*. (All data from USC genome browser.)

semblies, which makes them difficult to thoroughly compare. Further, results can vary according to the analysis methods selected. We emphasize that this is a preliminary analysis valuable for speculation and pointing to aspects of genome architecture that might merit further attention.

There are various sources which can be utilized for the human/opossum homology from the original genome assembly publication [Mikkelsen et al., 2007] to that of Wang et al. [2011]. We also assembled an electronic painting of the opossum/human homologies (online suppl. fig. 1; for all online supplementary material, see www.karger.com/doi/10.1159/000341502) based on a manual inspection of coding genes (USC genome browser, http://genome.ucsc.edu/cgi-bin/hgGateway). Marilyn Renfree

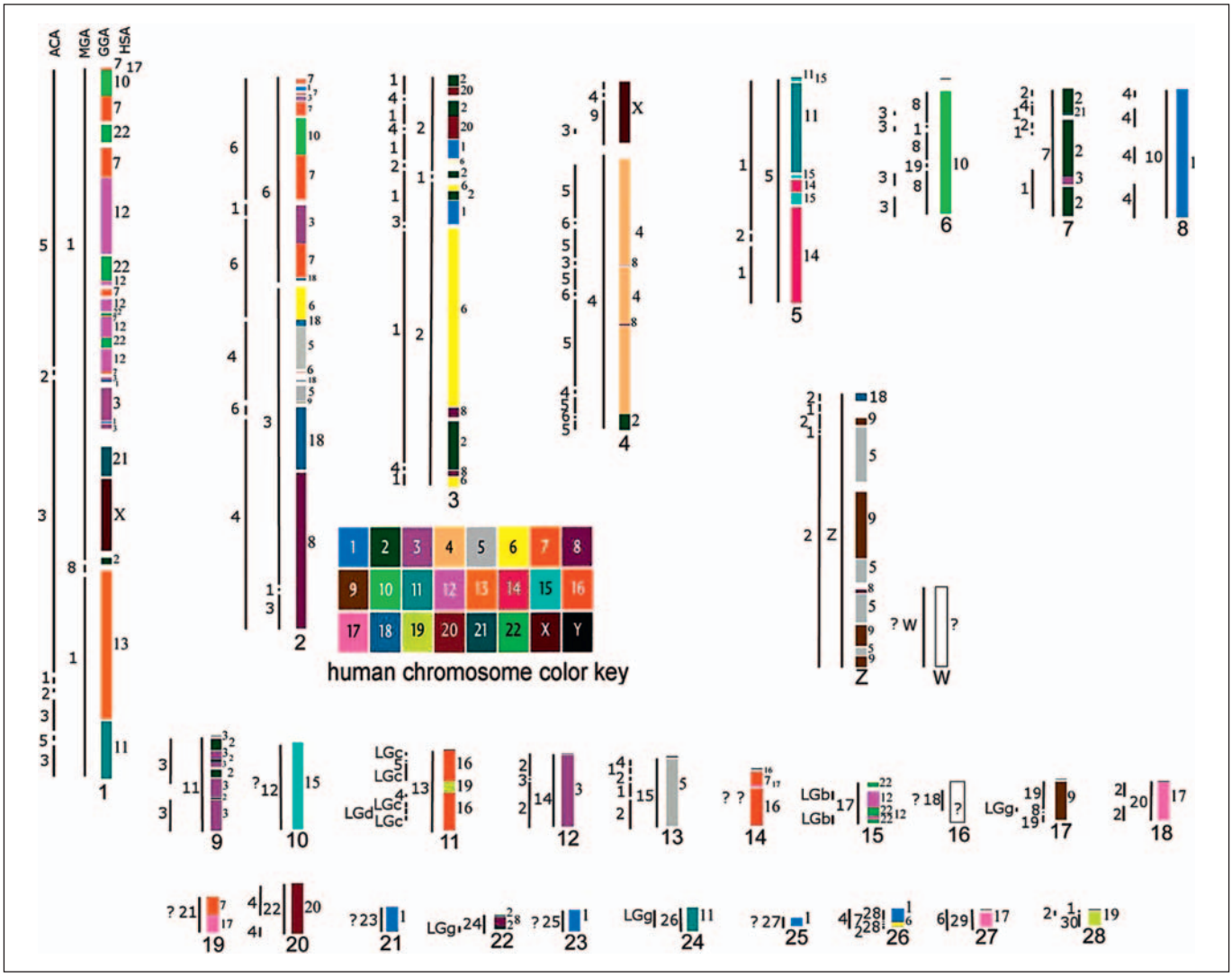

Fig. 3. Electronic chromosome painting idiogram of the chicken (GGA) showing the color coded homology with human chromosomes. The idiogram was produced by a manual examination of coding gene homology with the human genome. Chromosome segments between coding genes homologous to the same human chromosome were considered to be homologous to that human chromosome. Segmental homology <3 kb was ignored as it was considered to be under the resolution of chromosomes paints. The homologous regions to anole lizard (ACA) and turkey (MGA) were taken from a manual examination of scaffolds. (All data from USC genome browser.)

was kind enough to send M.A.F.-S. a human/tammar wallaby homology map, which was used for comparison. We also made an electronic painting karyotype of the chicken (fig. 3) for comparison to the other genomes based on a manual inspection of coding genes (USC genome browser). Comparisons with the lizard genome [Alfoldi et al., 2011] were made by taking the association data from the scaffolds of the genome assembly (USC genome browser). For all these comparisons we considered only major segmental associations which could possibly be seen at the cytogenetic level. We then listed associations which were found in at least 2 species (online suppl. table 1).

Among these vertebrates the highest affinity, as expected, is between tammar wallaby and opossum: there were 37 common associations: 1/6, 1/11, 1/17, 1/19, 2/3, 2/8, 2/11, 2/13, 2/15, 2/16, 2/20, 3/7, 3/9, 3/X, 4/8, 4/11, 5/14, 5/17, 5/18, 5/20, 6/8, 6/17, 6/18, 7/9, 7/10, 8/18, 8/19, 9/16, 9/17, 9/20, 10/15, 11/13, 12/22, 14/15, 15/X, 16/19, 21/X. It is interesting to note that the tammar wallaby, opossum

and platypus share 10 associations: 2/3, 2/13, 2/16, 2/20, 3/9, 5/18, 6/17, 7/10, 7/16, 8/18. Therefore, the opossum and tammar wallaby share 27 associations not found in the platypus.

In the same manner there are a few associations held by MDO and other vertebrates but not MEU: 2/5, 5/9, 10/12, and conversely there are also associations held by MEU with another vertebrate but not with MDO: 2/21, 7/12. It is informative to examine the associations held in common between non-placental mammals and other vertebrates. The 49 associations held in common between at least one mammal and the chicken or green anole lizard are compared in online supplement table 1. Of these 49 associations only 7 were apparently conserved in the ancestral eutherian karyotype.

Certainly, not all these associations are homologous and some may be due to convergence. However, the implications seem inescapable. It has long been appreciated that the platypus genome has a notable similarity with birds and reptiles [Rens et al., 2007; O'Brien, 2008; Veyrunes et al., 2008; Warren et al., 2008; Alfoldi et al., 2011]. We show here that this similarity extends to the common syntenic associations between platypus, birds and lizard. Perhaps surprisingly, there is also by far a greater similarity in association structure between opossum/tammar wallaby with birds and lizards than with placental mammals. Therefore, it does not seem to be an exaggeration to suggest that placentals have had a virtual genomic revolution at least at the association level. If the speculations of M.A.F.-S. presented earlier hold out, then there was a parallel revolution in the non-coding part of the genome. Perhaps we can further speculate that we have something entirely new in the chromosome makeup of placentals. In this regard, previously workers, to address the ancestral genome structure of eutherians, were apparently more interested in looking at the associations, which were held in common between marsupials and placentals. We suggest that it is equally important, and perhaps even more informative, to look at the ancestral vertebrate associations that are conserved in marsupials and monotremes but lost in placentals. The loss of the majority of ancient vertebrate associations helps us better appreciate the magnitude of what happened at the origin of placentals. On the basis of the association data we find no support for the suggestion of Ruiz-Herrera et al. [2012] that the placental ancestral karyotype is ancestral for all mammalian species. Further, we have to note that we were not able to utilize their homology maps of some of these species because of an unfortunate, inconsistent use of color-coding to human chromosome homology.

Future Directions in Molecular Cytogenetics

Although we now have, as this volume amply illustrates, a substantial, broad view of chromosome evolution in mammals, there are clearly many glaring gaps, in particular the insufficiency of taxa sampling. Phylogenomic analysis is better the more taxonomically rich are the assemblages available for analysis. Even in the best-studied mammalian orders, primates and carnivores, only a minority of species have been studied. Many contributors to the present special issue remark on how it would be beneficial to study additional species. This lack is particularly clear as pointed out by authors of papers in this special issue such as Volleth and Eick for bats and Romanenko and Volobouev for rodents. For both these enormous orders we have only just begun to scratch the surface of the chromosome diversity present. Additional species would also be helpful all over the mammalian tree and even in smaller clades such as Afrotheria and Xenarthra. We are sure that cytogeneticists will continue to fill in the missing species, taking full advantage of the progress that has been made up to this point. Indeed, several contributions to this review issue actually contributed new original data. We can mention that, among others, Biltueva and Vorobieva added chromosome painting data on Iberian shrew and Altai mole, while Perelman et al. report on chromosome painting of *Genetta pardina*.

Clearly the most widely utilized molecular cytogenetic method is chromosome painting. Very few species, most of them primates, have been studied with higher resolution methods such as BAC-FISH. Studies to define marker order (intrachromosomal rearrangements) and to describe inversions etc. by defining breakpoints via BAC-FISH need to be applied to a wider range of species. There is much work to be done at this level.

Nonetheless, one of the strengths of chromosome painting versus sequencing analyses is that the economy of cytogenetic analysis has allowed a larger, more appropriate sampling of taxa [Froenicke et al., 2006; Rocchi et al., 2006]. Instead, up to now, sequencing efforts have been devoted mostly to species of economic interest or to those that are of particular significance for understanding human evolution such as the higher primates. However, clearly sequencing efforts are rapidly widening their scope and, as costs drop, the number of vertebrate species that will be sequenced will be increasing exponentially. Consequently, cytogenetics will be called upon to aid sequencing efforts and can take advantage of the detailed information coming from these studies.

Integration of Cytogenetics and Genomic Sequencing

The integration of cytogenetic and sequencing information will be vital to helping resolve long-standing issues in cytogenetics, many of which are discussed in the contributions to this issue. Among these is resolving problems in reconstructing ancestral karyotypes at various points of the mammalian tree. Volleth and Eick discuss how Robertsonian translocations have obscured the karyotype composition of the chiropteran ancestor. The problem of weighing cytogenetic traits in phylogenomic analysis remains a point of discussion without clear resolution. For instance, the dangers of confusing convergence with homology exist even in the chromosome painting era. It is helpful to know the gene content especially at junctures of segments. A number of authors in this special issue attempt to use and integrate other types of biomolecular data, in part to eliminate the homoplasy that can plague cytogenetic reconstructions. Perelman et al. in carnivores compared results of reciprocal human-dog painting data and syntenic blocks order identified in Ensemble and in Evolutionary Highway. Trifonov et al. used genome sequencing data to test the homology of associations in Perissodactyla. Nie followed a similar procedure to refute the proposal of Picone et al. [2011] that Dermoptera and Lagomorpha formed a monophyletic clade supported by 3 synapomorphies, i.e. HSA4/18, 7/19 and 1/10, by showing that these associations are due to convergence.

As these authors showed, convergent association and homologous associations can sometimes be distinguished by reference to sequencing data since the contiguous segments should have the same gene content and order. Other cases of homoplasy such as those due to lineage sorting and reticulate evolution will not be ruled out by this method. So a combination of cytogenetic and sequence analysis could provide precious information on speciation processes.

Rates of Chromosome Evolution

Early painting comparisons of humans and other primates showed that there is no molecular clock for chromosome evolution. In other words, evolutionary rates in chromosome rearrangements vary greatly from one evolutionary line to another. In this special issue a number of contributions present examples of contrasting rates of chromosome evolution. Perelman et al. discuss both the high chromosome conservation in felids and the rapid chromosome evolution in canids and skunks. They also provide new data on another carnivore species with highly rearranged karyotype: the pardine genet. Rubes et al. discuss the conserved karyotypes of whales versus the highly rearranged ones of pigs. Trifonov et al. show how varied the evolutionary rates are for Perissodactyla, with extremely fast rates of chromosome evolution and centromere formation in equids. Richard and Dutrillaux reported on the 2 groups Sciuromorpha and Lagomorpha, normally known for their low number of rearrangements, but even here there are exceptions. De Oliviera et al. detail the remarkable example of chromosome variability in New World monkeys marked by vastly different rates of chromosome evolution. They contrast and compare various hypotheses of phylogenomics based on chromosome painting with reconstructions coming from other biomolecular comparisons.

There seems to be little room for the old concept that mammals have a default chromosome rearrangement rate with just a few exceptions. There are just too many exceptions and just too wide a range to accept that there is a default rate. However, the reasons for the differences have eluded our understanding. As discussed in the contribution by Stanyon et al., numerous hypotheses have been proposed including social organization, demographic parameters, meiotic drive, recombination suppression, as well as chromosome architecture such as segmental duplications and repeats.

In addition to their well-known and extremely rapid chromosome evolution, Romanenko and Volobouev detail the special genomic features of rodents. Certainly one of the most remarkable features of rodents is not only their outstanding variability in chromosome numbers and/or chromosome morphology, but also their heterogeneous heterochromatin, B and sex chromosomes. Such variability is present in other mammals, but perhaps not so dramatically.

It is probable that a comprehensive understanding of the forces responsible for varying rates of chromosome evolution and special characteristics of the mammalian genome will be at least in part found at the sequencing level in the coming years. A closer integration between cytogenetics and sequencing efforts will certainly be necessary if cytogeneticists want to maintain their relevance in evolutionary studies. One encouraging trend seen in the papers of this volume is the willingness of researchers to integrate and compare the cytogenetic data with phylogenomic information that comes from a wide array of biomolecular studies. We believe that the contributions here in no small way point the way to the next phase in cytogenetic studies.

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